Problems In Group Theory

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Contents

Chapter 1

Basic Group Theory

1.1 Introduction

Problem 1.1.1. *Determine which of the following binary operations are associative:*

1. *the operation on $\mathbb{Z}$ defined by $a * b = a - b$*
2. *the operation on $\mathbb{R}$ defined by $a * b = a + b + ab$*
3. *the operation on $\mathbb{Q}$ defined by $a * b = \frac{a+b}{5}$*
4. *the operation on $\mathbb{Z} \times \mathbb{Z}$ defined by $(a, b) * (c, d) = (ad + bc, bd)$*
5. *the operation on $\mathbb{Q} - \{0\}$ defined by $a * b = \frac{a}{b}$*

Proof. This exercise is pretty trivial as we just have to compute $a * (b * c) = (a * b) * c$. For 1., $a * (b * c) = a * (b - c) = a - (b - c) = a - b + c$ but $(a * b) * c = (a - b) * c = (a - b) - c = a - b - c$. Rest of them are pretty similar. $\square$